THE MYSTERIOUS CASE OF ATTRITION

Exploratory workforce analytics | Python, SQL and data storytelling

Business question
What patterns distinguish employees who left from those who stayed, and what should an organisation investigate before deciding how to improve retention?

Metric	Result
Employees in dataset	1,470
Employees who left	237
Overall attrition rate	16.12%
Highest job-role attrition rate	Sales Representative — 39.76% (33 of 83)
Overtime group attrition rate	30.53% with overtime vs 10.44% without

Executive summary

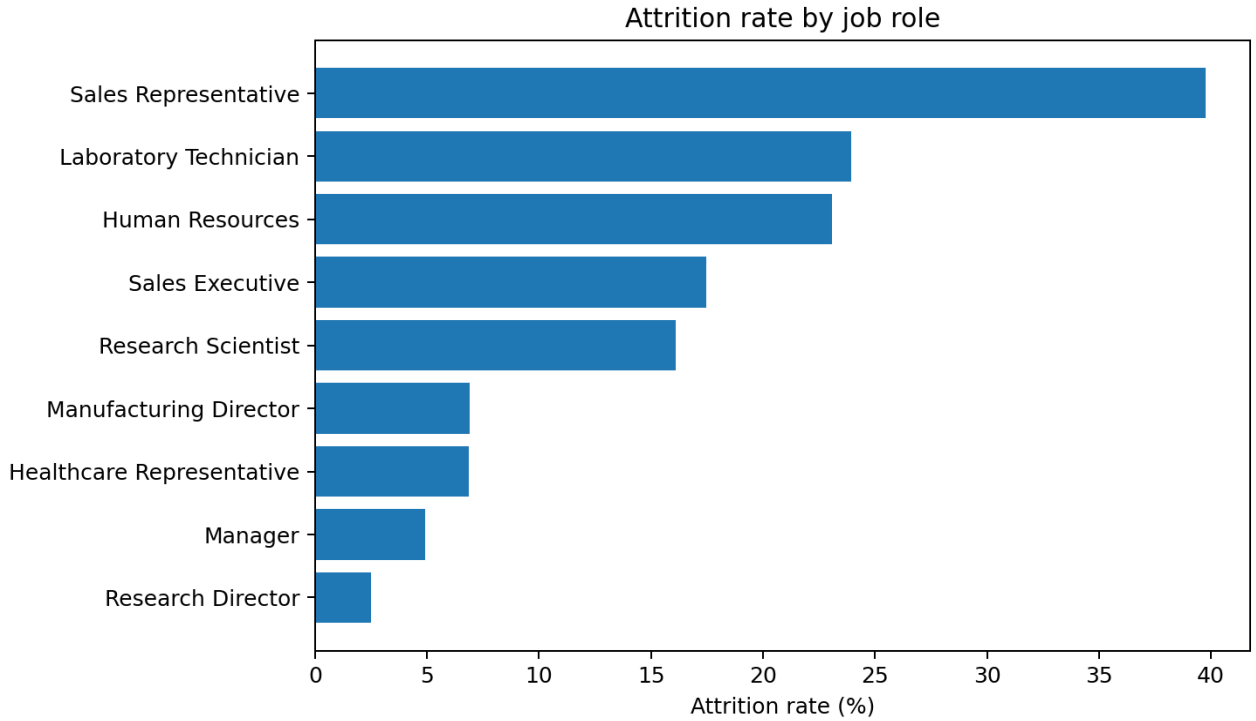
In this exploratory analysis, attrition varies across job roles, overtime status, tenure and employee-experience scores. Sales Representatives have the highest observed job-role attrition rate. Employees recorded as working overtime have a higher attrition rate than employees not working overtime. Attrition is also higher in the dataset's early-tenure groups and among employees with the lowest environment-satisfaction score.

Interpretation: These patterns identify areas worth investigating; they do not establish why employees left or prove that any factor caused attrition. The dataset is synthetic, so the results should be treated as a demonstration of analytical method rather than findings about a real employer.

Dataset: IBM HR Analytics Employee Attrition & Performance (synthetic dataset).

1. Attrition differs by job role

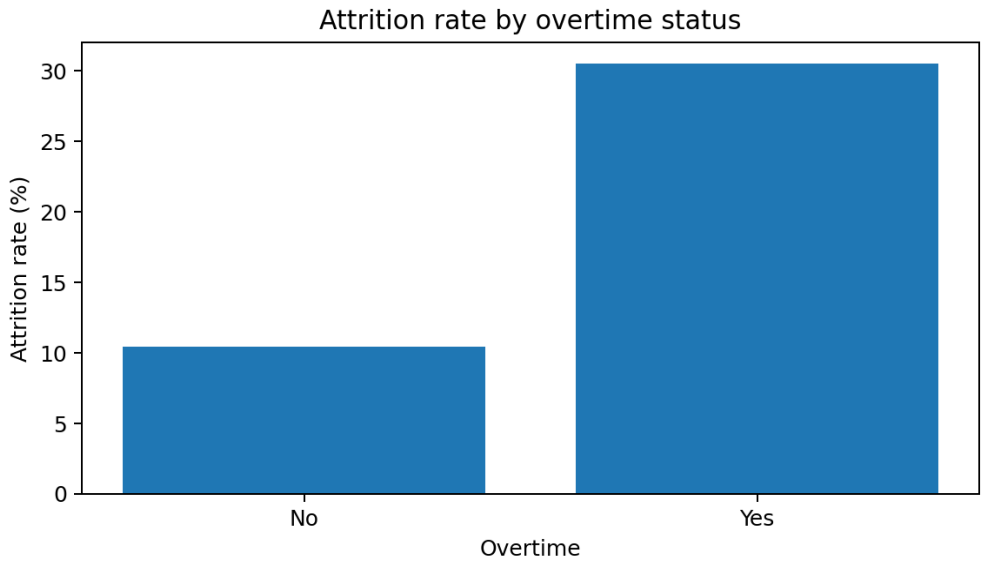
The highest observed rate is among Sales Representatives (39.76%, 33 of 83 employees). Laboratory Technicians have the second-highest rate in this dataset (23.94%, 62 of 259). The role with the highest rate is not necessarily the role with the most departures: Laboratory Technicians have 62 departures, while Sales Representatives have 33.



Job role	Employees	Left	Rate
Sales Representative	83	33	39.76%
Laboratory Technician	259	62	23.94%
Human Resources	52	12	23.08%
Sales Executive	326	57	17.48%
Research Scientist	292	47	16.10%
Manufacturing Director	145	10	6.90%
Healthcare Representative	131	9	6.87%
Manager	102	5	4.90%
Research Director	80	2	2.50%

Question to investigate: Are role expectations, staffing, compensation, workload or manager support different in these roles? The dataset cannot answer those questions directly.

2. Overtime and early tenure



Employees working overtime have a 30.53% attrition rate (127 of 416), compared with 10.44% (110 of 1,054) among employees not working overtime. This is approximately 2.92 times the observed rate, a difference of 20.09 percentage points.

Early-tenure comparison

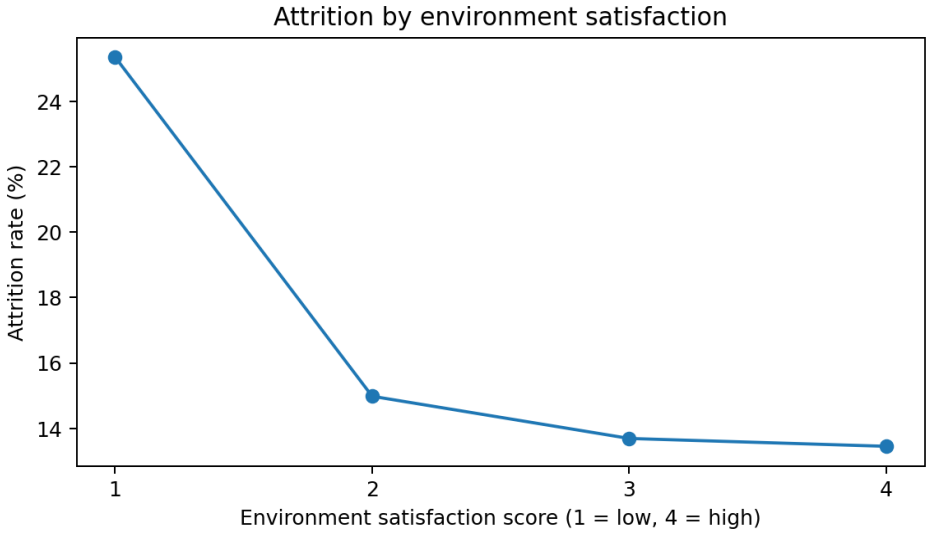
Years at company	Employees	Left	Attrition rate
0 years	44	16	36.36%
1 year	171	59	34.50%
0–1 years combined	215	75	34.88%
5 years	196	21	10.71%

Among employees with 0–1 years at the company, the attrition rate is 55.07% for those working overtime (38 of 69) and 25.34% for those not working overtime (37 of 146). This subgroup comparison is descriptive; role mix and other factors may contribute to the difference.

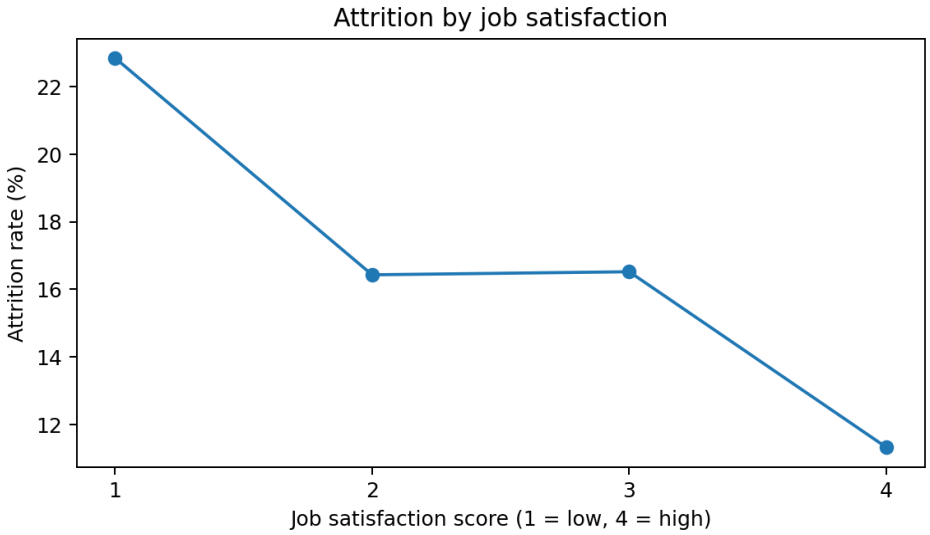
Questions to investigate: Are overtime patterns sustainable? Do new employees receive clear role expectations, appropriate onboarding and enough manager support?

3. Employee-experience scores

Lower satisfaction scores are associated with higher observed attrition in several comparisons. These scores do not explain why a person left, and the patterns are not proof that changing a score will reduce attrition.



Environment satisfaction: the attrition rate is 25.35% at score 1 (72 of 284 employees) and 13.45% at score 4 (60 of 446 employees).



Job satisfaction: the rate is 22.84% at score 1 and 11.33% at score 4. Scores 2 and 3 are similar (16.43% and 16.52%).

Work-life balance has a less linear pattern: score 1 has a 31.25% rate, score 3 has 14.22%, and score 4 has 17.65%. The score-1 group contains 80 employees, so interpret the percentage cautiously.

4. Recommended next investigations

Review role-level conditions. Investigate job expectations, staffing, workload, compensation structures and manager support in roles with higher observed attrition.

Examine overtime practices. Review overtime frequency, distribution across teams and whether workload or staffing constraints may be relevant.

Strengthen early-tenure listening. Use onboarding check-ins and employee feedback to test whether role clarity, training and manager support are relevant retention factors.

Explore employee experience. Pair satisfaction scores with qualitative feedback and real workforce data before selecting interventions.

Validate with real data. Repeat the analysis on an appropriate real-world dataset and assess whether the patterns persist before making business decisions.

Method and limitations

• The dataset is synthetic and does not represent actual employees at a real organisation.

• This is descriptive analysis; it does not establish causation or identify the reason an employee left.

• Some subgroups are small. For example, the early-tenure Human Resources rate is 80% but represents only 4 departures among 5 employees.

• Rates and counts should be considered together. A high rate in a small group can be unstable, while a larger group may contribute more departures.

• The analysis uses the supplied dataset's recorded categories and scores; it does not validate how those variables were collected or defined.

Technical workflow

Python/pandas was used for data inspection and exploratory analysis. SQLite was used to independently validate overall attrition, job-role attrition and overtime attrition. Matplotlib charts communicate the main patterns.

Data source: IBM HR Analytics Employee Attrition & Performance — Kaggle.